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AI-DRIVEN CYBER SECURITY

YOUR AI APPLICATION IS RUNNING PERFECTLY, BUT HOW DO YOU KNOW IT'S ACTUALLY GIVING CORRECT ANSWERS?

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AI OBSERVABILITY

- AI Observability means understanding what's happening inside an AI application.
- It helps us monitor prompts, responses, errors, latency, token usage, and model behavior.
- It helps identify why an AI system produces unexpected results.
- Think of it as a monitoring system for AI applications.

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AI LOGS

- Logs are records of events happening inside an AI application.
- They can capture user prompts, model responses, errors, timestamps, and execution details.
- Logs help developers investigate failures and debug AI behavior.
- Never log passwords, API keys, or sensitive personal information.

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PROMPT TRACING

- Prompt tracing tracks the complete journey of an AI request.
- It records which components process the request and how long each step takes.
- In a RAG system, we can trace retrieval, prompt construction, LLM generation, and final response.
- It helps us identify exactly where something goes wrong.

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LANGFUSE

- Langfuse is an open-source LLM observability platform.
- It helps track prompts, generations, traces, latency, token usage, and costs.
- It provides dashboards to inspect and evaluate AI application behavior.
- It is especially useful for debugging and improving LLM applications.

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OPENTELEMETRY

- OpenTelemetry is an open-source observability framework.
- It collects and exports telemetry data such as traces, metrics, and logs.
- It helps monitor different services in a distributed AI application.
- It is not an AI-specific evaluation tool; it provides the observability foundation.

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AI EVALUATION

- AI Evaluation measures how well an AI application performs.
- It checks accuracy, relevance, correctness, reliability, and other quality metrics.
- We can use predefined test datasets and expected answers.
- Evaluation helps us compare different prompts, models, and system versions.

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PRECISION

- PRECISION MEASURES HOW MANY DETECTED POSITIVE RESULTS ARE ACTUALLY CORRECT.
- HIGH PRECISION MEANS FEWER FALSE ALARMS.

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RECALL

- RECALL MEASURES HOW MANY ACTUAL POSITIVE CASES THE SYSTEM SUCCESSFULLY DETECTS.
- HIGH RECALL MEANS FEWER MISSED PROBLEMS.

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FALSE POSITIVES & FALSE NEGATIVES

- FALSE POSITIVE: THE SYSTEM FLAGS SOMETHING AS WRONG OR UNSAFE WHEN IT IS ACTUALLY CORRECT OR SAFE.
- FALSE NEGATIVE: THE SYSTEM FAILS TO DETECT SOMETHING THAT IS ACTUALLY WRONG OR UNSAFE.
- EXCESSIVE FALSE POSITIVES FRUSTRATE USERS.
- FALSE NEGATIVES ALLOW DANGEROUS OR INCORRECT OUTPUTS TO PASS.

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AUTOMATED HALLUCINATION DETECTION

- A HALLUCINATION OCCURS WHEN AN AI GENERATES INCORRECT OR UNSUPPORTED INFORMATION.
- AUTOMATED DETECTION CHECKS WHETHER AN AI ANSWER IS SUPPORTED BY RELIABLE EVIDENCE.
- IN RAG SYSTEMS, WE CAN COMPARE GENERATED ANSWERS AGAINST RETRIEVED DOCUMENTS.
- EVALUATION METHODS CAN MEASURE FACTUAL CONSISTENCY, RELEVANCE, AND GROUNDEDNESS.
- DETECTION TOOLS CAN MAKE MISTAKES, SO HUMAN REVIEW IS IMPORTANT FOR CRITICAL DECISIONS.

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REDUCING FALSE POSITIVES

- USE HIGH-QUALITY EVALUATION DATASETS.
- SET APPROPRIATE DETECTION THRESHOLDS.
- COMBINE MULTIPLE EVALUATION SIGNALS INSTEAD OF RELYING ON ONE RULE.
- TEST ON BOTH VALID AND MALICIOUS EXAMPLES.
- REVIEW FALSE ALARMS AND CONTINUOUSLY IMPROVE DETECTION LOGIC.